

Reactions of the Period 3 Elements (Cambridge (CIE) A Level Chemistry): Revision Note

Reactions of the Period 3 Elements

Reactions with oxygen & chlorine

Reaction of Period 3 elements with oxygen table

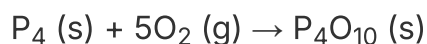
	Chemical equation	Reaction conditions	Reaction	Flame	Product
Na	$4\text{Na (s)} + \text{O}_2 \text{ (g)} \rightarrow 2\text{Na}_2\text{O (s)}$	Heat	Vigorous	Bright yellow	White solid
Mg	$2\text{Mg (s)} + \text{O}_2 \text{ (g)} \rightarrow 2\text{MgO (s)}$	Heat	Vigorous	Bright white	White solid
Al	$4\text{Al (s)} + 3\text{O}_2 \text{ (g)} \rightarrow 2\text{Al}_2\text{O}_3 \text{ (s)}$	Powdered Al	Fast	Bright white	White powder
Si	$\text{Si (s)} + \text{O}_2 \text{ (g)} \rightarrow \text{SiO}_2 \text{ (s)}$	Powdered Si Heat strongly	Slow	Bright white sparkles	White powder
P	$4\text{P (s)} + 5\text{O}_2 \text{ (g)} \rightarrow \text{P}_4\text{O}_{10} \text{ (s)}$	Heat	Vigorous	Yellow or white	White clouds
S	$\text{S (s)} + \text{O}_2 \text{ (g)} \rightarrow \text{SO}_2 \text{ (g)}$	Powdered S Heat	Gentle	Blue	Toxic fumes





Examiner Tips and Tricks

Phosphorus typically exists as P_4 , which means that the equation for the oxidation could also be represented as:

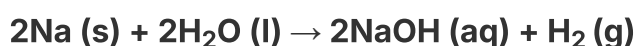


Reaction of Period 3 elements with chlorine table

	Chemical equation	Reaction conditions	Reaction
Na	$2Na (s) + Cl_2 (g) \rightarrow 2NaCl (s)$	Heat	Vigorous
Mg	$Mg (s) + Cl_2 (g) \rightarrow MgCl_2 (s)$	Heat	Vigorous
Al	$2Al (s) + 3Cl_2 (g) \rightarrow Al_2Cl_6 (s)$	Heat	Vigorous
Si	$Si (s) + 2Cl_2 (g) \rightarrow SiCl_4 (l)$	Heat	Slow
P	$2P (s) + 5Cl_2 (g) \rightarrow 2PCl_5 (l)$	Heat Excess chlorine	Slow

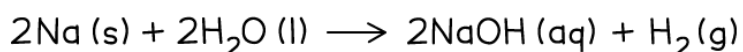
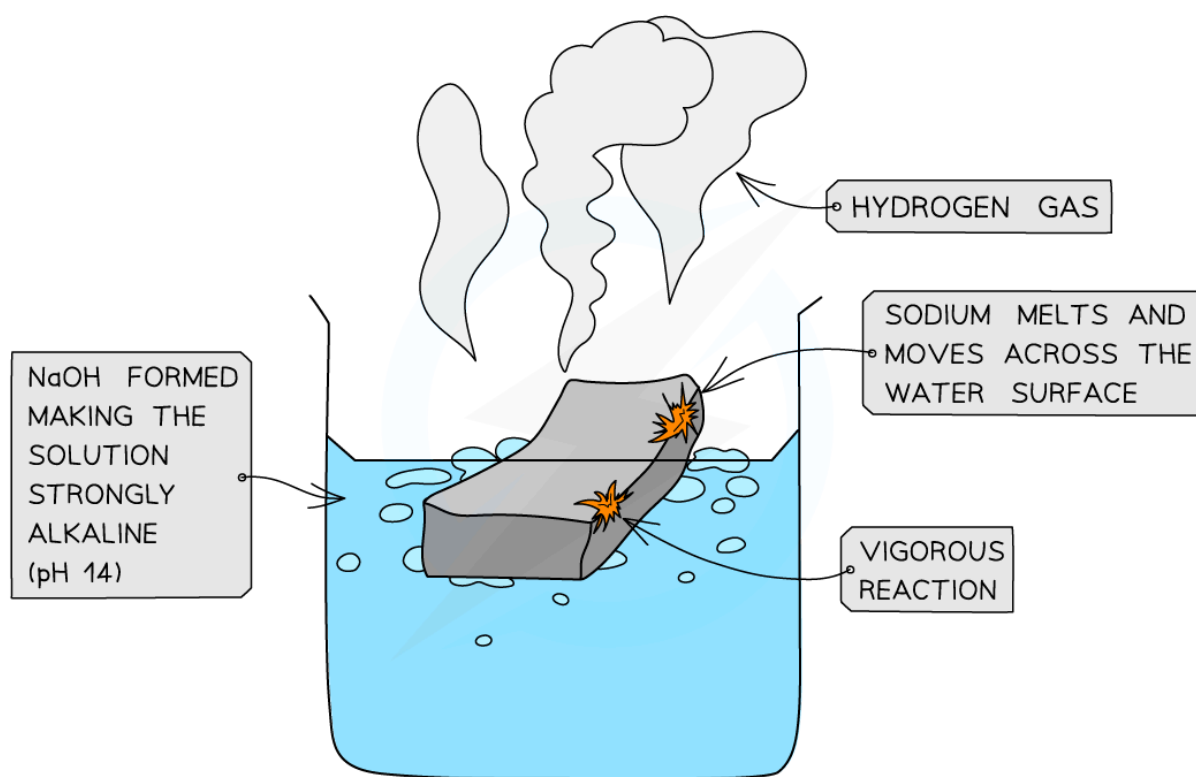
Reaction of sodium with water

- Sodium reacts **vigorously** with **cold water**:



- The sodium **melts** into a ball and moves across the water surface until it disappears
- Hydrogen gas** is given off
- The solution formed is **strongly alkaline** (pH 14) due to the **sodium hydroxide** which is formed

The reaction of sodium with cold water



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Sodium fizzes while vigorously reacting with water to form sodium hydroxide

Reaction of magnesium with water

- Magnesium reacts **extremely slowly** with cold water:



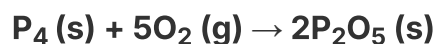
- The solution formed is **weakly alkaline** (pH 11) as the formed **magnesium hydroxide** is only **slightly soluble**
- When magnesium is **heated**, it reacts **vigorously** with steam (water) to make magnesium oxide and hydrogen gas:



Examiner Tips and Tricks

Phosphorus exists as an allotrope. A common allotrope is P₄ and is known as white phosphorus. It can also exist as red phosphorus.

White phosphorus reacts spontaneously with **excess** oxygen to produce phosphorus(V) oxide via the following reaction.



Oxidation Number of Period 3 Oxides & Chlorides

- Chlorine is more **electronegative** than the other Period 3 elements
 - Therefore, the other Period 3 elements will generally have **positive oxidation states** in their chlorides, while the chlorine has a **negative oxidation state** of -1
- Oxygen is more **electronegative** than any of the Period 3 elements
 - Therefore, the Period 3 elements will have **positive oxidation states** in their oxides, while the oxygen has a **negative oxidation state** of -2

The Pauling scale of electronegativity

PAULING ELECTRONEGATIVITY VALUES FOR THE ELEMENTS

H 2.1																	He –
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne –
Na 0.9	Mg 1.2											Al 1.5	Si 1.8	P 2.2	S 2.5	Cl 3.0	Ar –
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr 3.0
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5	Xe 2.6
Cs 0.7	Ba 0.9	La–Lu 1.1–1.2	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2	Rn –
Fr 0.7	Ra 0.9	Ac–No 1.1–1.7															

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The Pauling scale shows that chlorine has a higher electronegativity than the other Period 3 elements and oxygen has a higher electronegativity than all of the Period 3 elements

Oxidation numbers of Period 3 elements in their oxides

- The normal [Oxidation Number Rules](#) apply to the Period 3 oxides

Table of formulae of the Period 3 oxides & the elements oxidation numbers

Period 3 element	Na	Mg	Al	Si	P	S
Formula of oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₂ SO ₃
Oxidation number of the Period 3 element	+1	+2	+3	+4	+5	+4 +6



Examiner Tips and Tricks

- SiO₂ is not written in the specification as expected knowledge but you should be able to apply the oxidation number rules to this and other chemicals



Worked Example

Deducing oxidation numbers of Period 3 elements in their oxides

State the oxidation number of the bold atoms in these compounds or ions.

1. **Na**₂O
2. **Al**₂O₃
3. **P**₄O₁₀

Answer

1. **Na**₂O

- 1 O atom = -2
- The overall charge of the compound = 0
- 2 Na atoms = +2
- Oxidation number of 1 Na atom = $(+2) / 2 = +1$

2. **Al₂O₃**

- 3 O atoms = $3 \times (-2) = -6$
- The overall charge of the compound = 0
- 2 Al atoms = +6
- Oxidation number of 1 Al atom = $(+6) / 2 = +3$

3. **P₄O₁₀**

- 10 O atoms = $10 \times (-2) = -20$
- The overall charge of the compound = 0
- 4 P atoms = +20
- Oxidation number of 1 Al atom = $(+20) / 4 = +5$

Oxidation numbers of Period 3 elements in their chlorides

- The normal [Oxidation Number Rules](#) apply to the Period 3 chlorides

Table of formulae of the Period 3 chlorides & the elements oxidation numbers

Period 3 element	Na	Mg	Al	Si	P
Formula of chloride	NaCl	MgCl ₂	AlCl ₃	SiCl ₄	PCl ₅
Oxidation number of the Period 3 element	+1	+2	+3	+4	+5



Worked Example

Deducing oxidation numbers of Period 3 elements in their chlorides

State the oxidation number of the bold atoms in these compounds or ions.

1. Na_2O
2. Al_2O_3
3. P_4O_{10}

Answer

1. MgCl_2
 - 2 Cl atoms = $2 \times (-1) = -2$
 - The overall charge of the compound = 0
 - Oxidation number of 1 Mg atom = +2
 - Or, the oxidation number of a Group 2 metal is +2
2. AlCl_3
 - 3 Cl atoms = $3 \times (-1) = -3$
 - The overall charge of the compound = 0
 - Oxidation number of 1 Al atom = +3
3. PCl_5
 - 5 Cl atoms = $5 \times (-1) = -5$
 - The overall charge of the compound = 0
 - Oxidation number of 1 P atom = +5

Reaction of Period 3 Oxides & Water

- Not all Period 3 oxides **react** with or are **soluble** in water

Reaction of Period 3 oxides with water table

Oxide	Chemical equation	pH	Comments
Na_2O	$\text{Na}_2\text{O (s)} + \text{H}_2\text{O (l)} \rightarrow 2\text{NaOH (aq)}$	12 - 14 (Strongly alkaline)	-
MgO	$\text{MgO (s)} + \text{H}_2\text{O (l)} \rightarrow \text{Mg(OH)}_2 \text{ (aq)}$	8 - 10 (Weakly alkaline)	-
Al_2O_3	No reaction	-	Al_2O_3 is insoluble in water

SiO₂	No reaction	-	SiO ₂ is insoluble in water
P₄O₁₀	$\text{P}_4\text{O}_{10} (\text{s}) + 6\text{H}_2\text{O} (\text{l}) \rightarrow 4\text{H}_3\text{PO}_4 (\text{aq})$	3 - 4 (Weakly acidic)	Vigorous reaction
SO₂	$\text{SO}_2 (\text{s}) + \text{H}_2\text{O} (\text{l}) \rightleftharpoons \text{H}_2\text{SO}_3 (\text{aq})$	1 - 2 (Strongly acidic)	-
SO₃	$\text{SO}_3 (\text{s}) + \text{H}_2\text{O} (\text{l}) \rightarrow \text{H}_2\text{SO}_4 (\text{aq})$	1 - 2 (Strongly acidic)	-



Examiner Tips and Tricks

- Since aluminium oxide does not react or dissolve in water, the oxide layer protects the aluminium metal from corrosion
- The reaction of silicon(IV) oxide is not required knowledge